

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1710

COURSE OUTCOME COVERED

CO5: *Design AI-based systems using PySpark, RDD operations, and intelligent agent architectures, using scalable systems and integrate components in distributed AI applications. (BTL 5)*

Assessment Tool Weightage: 20%

QUESTIONS: 5

Duration: 1 Hour
Total Marks:50

Annexure A

Reverse Engineering

Code of Conduct:

I Shaik Arshad Basha (192312336) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Sk Arshad Basha

Reverse Engineering an Airline Dynamic Pricing System

An airline adjusts ticket prices in real time based on demand, competitor pricing, and booking patterns across thousands of routes. Reverse engineer the underlying AI system by describing the data pipeline using RDD-based streaming of booking data, identifying agent interactions across routes, and explaining how pricing decisions are made under real-time market constraints.

Reverse Engineering an Airline Dynamic Pricing System

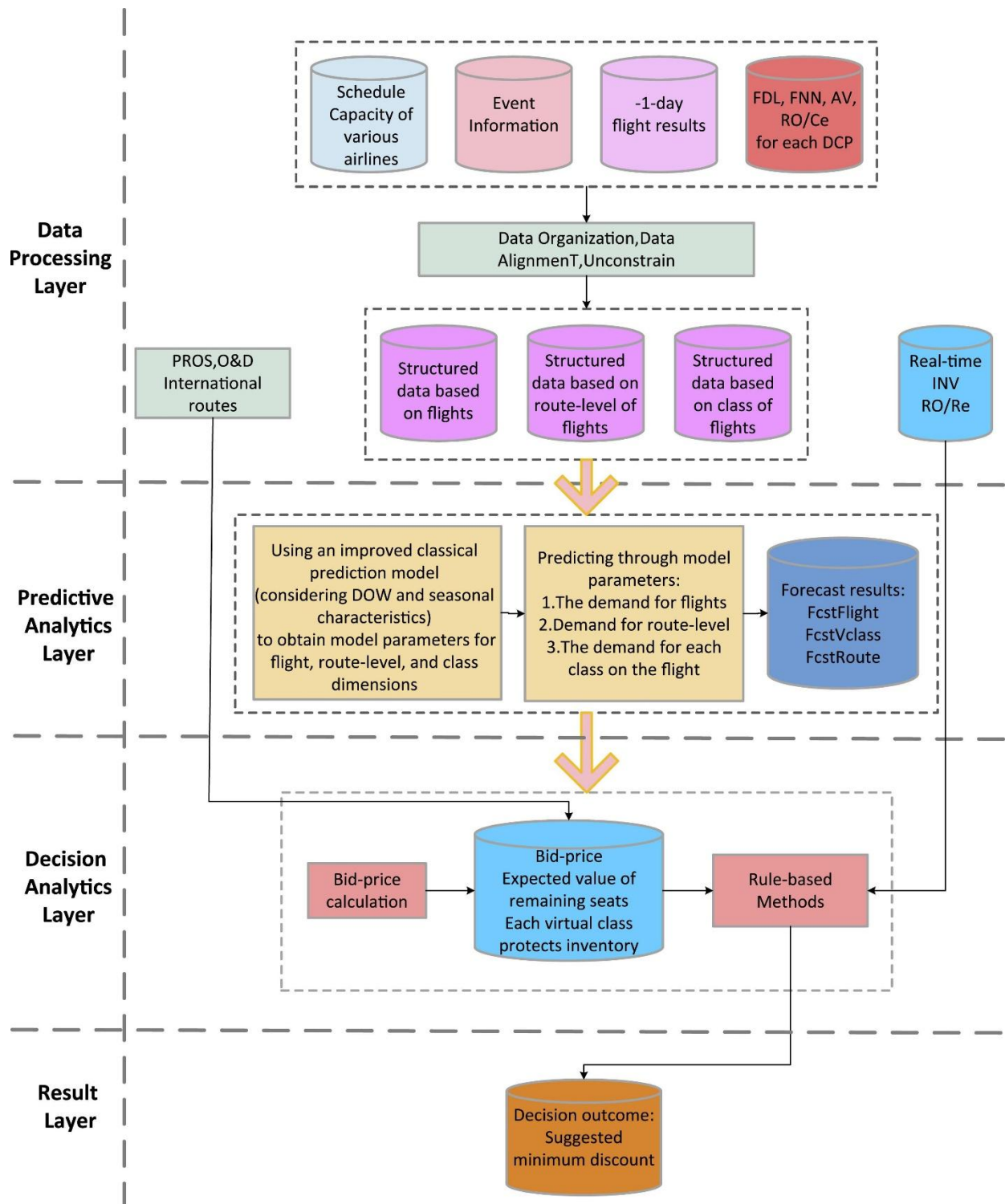
Introduction

An airline dynamic pricing system is an AI-driven decision system that continuously changes ticket prices according to customer demand, remaining seat capacity, booking patterns, competitor prices, travel dates, and market conditions.

Main Objective

- To understand how real-time airline booking data can be processed using Apache Spark RDDs.
- To identify how intelligent agents coordinate pricing decisions across thousands of routes.
- To explain how demand, competition, and booking behavior influence ticket prices.
- To model how the system makes pricing decisions under real-time constraints.

Overall System Architecture



Data Pipeline Using RDD-Based Streaming

The system receives continuous booking events from thousands of routes.

A booking record could contain:

Route = Chennai → Delhi

Flight = AI123

Date = 15-Sep

Seats Available = 42

Bookings = 158

Current Price = ₹7,500

Competitor Price = ₹7,200

Days to Departure = 10

These records are distributed across a Spark cluster.

Intelligent Agents

The system can be modeled as a multi-agent pricing system, where each agent specializes in one aspect of the decision.

1. Demand Agent

Monitors:

- Number of bookings
- Booking velocity
- Historical demand

- Seasonal patterns

Example:

Normal demand → Small price change

High demand → Price increase

Low demand → Price reduction/promotion

2. Capacity Agent

Monitors available seats.

Seats Available = 100



Bookings = 90



High Occupancy



Increase price pressure

When the aircraft is nearly full, the agent recommends a higher fare class or price.

3. Competitor Agent

Tracks competitor prices for comparable routes and travel dates.

Our Price = ₹7,500

Competitor A = ₹7,200

Competitor B = ₹7,400

The agent provides market information to the pricing engine.

Note: A production implementation should use only legally obtained and contractually permitted market data.

Pricing Decision Model

A simplified pricing model can be represented as:

$$P_{new} = P_{base} + \Delta P$$

where:

$$\Delta P = f(D, C, K, B, T)$$

Where:

- **D** = demand level
- **C** = available capacity
- **K** = competitor/market price
- **B** = booking behavior
- **T** = time remaining before departure

A conceptual scoring model could be:

$$Score = w_1D + w_2C + w_3K + w_4B + w_5T$$

The pricing engine converts this score into a permitted fare adjustment.

Real-Time Market Constraints

The pricing engine cannot simply choose any price. It must operate within constraints.

Important constraints

- Minimum and maximum permitted fare
- Available seats
- Fare-class inventory
- Time until departure
- Expected demand
- Competitor market conditions
- Promotional rules
- Regulatory and contractual requirements
- Revenue targets

Conclusion

- RDD-based distributed processing allows booking and market data from thousands of airline routes to be transformed, filtered, grouped, and aggregated in parallel.
- Multiple intelligent agents analyze demand, capacity, competitor prices, and booking patterns, while a pricing agent combines these signals to select a fare subject to real-time business and market constraints.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation